

Apollo 24/7 Buy Medicine Module Automation Testing Framework

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Project Type - Capstone Project

**Project Name - Automation Testing using Selenium PyTest Framework
& Automation Testing using Behave BDD Framework**

Domain - Apollo 247

Module Title - Buy Medicine

Synopsis

The Apollo 24/7 Buy Medicine Module Automation Testing Framework is a web automation project developed to automate and validate the core functionalities of the Apollo 24/7 medicine ordering application. The project focuses on improving software quality, reducing manual testing effort, and ensuring reliable validation of critical user workflows.

The framework is implemented using Python with Selenium WebDriver and follows the Page Object Model (POM) design pattern for better code reusability, scalability, and maintainability. Initially developed using Pytest, the framework was later enhanced into a Pure Behavior Driven Development (BDD) framework using Behave to support readable business-oriented test scenarios and structured automation execution.

The automation framework covers important functionalities such as:

- Website launch validation
- User login functionality
- Medicine search functionality
- Add to cart operations
- Checkout process validation
- Positive and negative test scenarios
- End-to-End user journey testing

The project also integrates:

- Allure Reports for advanced reporting and visualization
- CSV-based Data-Driven Testing for external test data management
- Screenshot capturing for execution evidence
- Logging mechanism for debugging and traceability

The framework generates detailed execution reports containing pass/fail status, screenshots, execution logs, and scenario-level results, making it highly useful for automation testing, regression testing, and continuous testing environments.

This project demonstrates practical implementation of modern automation testing concepts, BDD methodology, reporting integration, and scalable test framework design suitable for real-world industry applications.

Project Description

The Apollo 24/7 Buy Medicine Module Automation Framework is a web automation testing project developed using Selenium WebDriver with Python. The framework automates critical functionalities of the Apollo 24/7 medicine ordering workflow including website launch, login validation, medicine search, add to cart functionality, and checkout flow.

The project was initially developed using Pytest and later enhanced into a Pure BDD Framework using Behave for better readability, maintainability, and business-driven test execution.

The framework follows the Page Object Model (POM) design pattern and integrates Allure Reporting for advanced test execution visualization and reporting.

The framework supports:

- Data-driven testing using CSV files
- Positive and negative test validations
- End-to-End (E2E) workflow testing
- Screenshot capturing on test execution
- Detailed logging
- Allure reporting integration
- BDD feature-based execution

Test Strategy

Testing Approach

The project follows an Automation Testing Strategy based on:

- Functional Testing
- UI Testing
- Data-Driven Testing
- Behavior Driven Development (BDD)
- End-to-End Testing

Framework Design Pattern

The framework is implemented using:

- Page Object Model (POM)
- Modular architecture
- Reusable utility functions
- External test data management

Test Coverage

The automation framework validates the following functionalities:

Positive Test Scenarios

- Website launch verification
- Valid user login
- Valid medicine search
- Add medicine to cart
- Checkout process validation

Negative Test Scenarios

- Invalid login validation
- Invalid medicine search validation

End-to-End Test Scenarios

- Complete user journey from login to checkout

Data-Driven Testing

Test data is maintained externally using CSV files:

- login_data.csv
- medicine_search_data.csv
- product_validation_data.csv

This improves:

- Scalability
- Maintainability
- Reusability

Test Environment

Component	Details
Application Under Test	Apollo 24/7
Base URL	https://www.apollo247.com/
Automation Tool	Selenium WebDriver
Programming Language	Python
Testing Framework	Behave (BDD), Pytest
Design Pattern	Page Object Model (POM)
Reporting Tool	Allure Reports
Data Source	CSV Files
IDE Used	PyCharm
Browser	Google Chrome
Browser Driver	ChromeDriver
Operating System	Windows 11
Version Control	Git & GitHub

Test Execution

Execute BDD Test Scenarios

behave

Execute Specific E2E Scenario

behave --tags=@e2e

Generate Allure Report

allure generate reports/allure-results -o reports/allure-report --clean

Open Allure Report

allure open reports/allure-report

Execute Selenium (Pytest) Test Scenarios

pytest -v -s

Execute Specific E2E Scenario

pytest -s -v tests/test_e2e.py

Generate Allure Report

allure generate reports/allure-results -o reports/allure-report --clean

Open Allure Report

allure open reports/allure-report

Test Output

Successful Validations

The framework successfully validates:

- Apollo website accessibility
- Login functionality
- Medicine search functionality
- Cart operations
- Checkout navigation
- Negative input validations

Artifacts Generated During Execution

The framework automatically generates:

- Execution logs
- Screenshots
- Allure result files
- HTML reports

Logs Capture

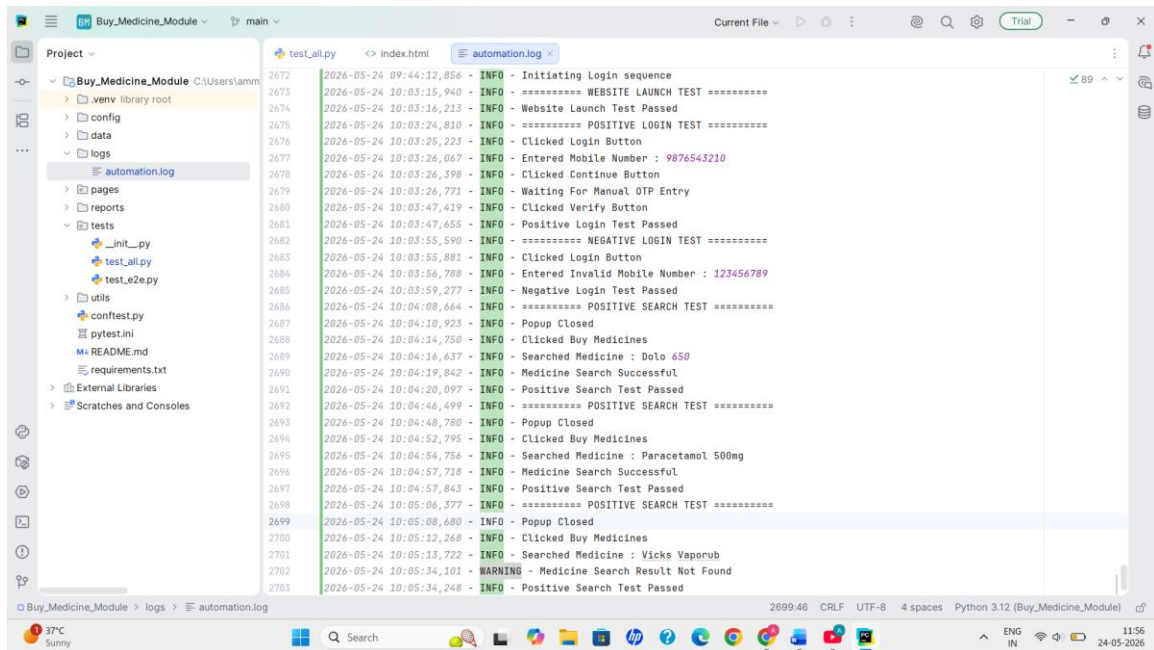
Screenshots are captured automatically:

- During execution
- On test completion
- On failures

```
2026-05-24 09:19:23,395 - INFO - ===== WEBSITE LAUNCH TEST =====
2026-05-24 09:19:23,990 - INFO - Website Launch Test Passed
2026-05-24 09:19:37,729 - INFO - ===== POSITIVE LOGIN TEST =====
2026-05-24 09:19:38,136 - INFO - Clicked Login Button
2026-05-24 09:19:39,102 - INFO - Entered Mobile Number : 9876543210
2026-05-24 09:19:39,385 - INFO - Clicked Continue Button
2026-05-24 09:19:39,669 - INFO - Waiting For Manual OTP Entry
2026-05-24 09:20:00,164 - INFO - Clicked Verify Button
2026-05-24 09:20:00,446 - INFO - Positive Login Test Passed
2026-05-24 09:20:11,277 - INFO - ===== NEGATIVE LOGIN TEST =====
2026-05-24 09:20:11,611 - INFO - Clicked Login Button
2026-05-24 09:20:12,145 - INFO - Entered Invalid Mobile Number : 123456789
2026-05-24 09:20:14,553 - INFO - Negative Login Test Passed
2026-05-24 09:20:23,684 - INFO - ===== POSITIVE SEARCH TEST =====
```

2026-05-24 09:20:25,920 - INFO - Popup Closed
2026-05-24 09:20:30,058 - INFO - Clicked Buy Medicines
2026-05-24 09:20:32,297 - INFO - Searched Medicine : Dolo 650
2026-05-24 09:20:35,259 - INFO - Positive Search Test Passed
2026-05-24 09:20:45,234 - INFO - ===== POSITIVE SEARCH TEST =====
2026-05-24 09:20:47,459 - INFO - Popup Closed
2026-05-24 09:20:49,019 - INFO - Clicked Buy Medicines
2026-05-24 09:20:51,028 - INFO - Searched Medicine : Paracetamol 500mg
2026-05-24 09:20:54,012 - INFO - Positive Search Test Passed
2026-05-24 09:21:03,105 - INFO - ===== POSITIVE SEARCH TEST =====
2026-05-24 09:21:05,425 - INFO - Popup Closed
2026-05-24 09:21:10,083 - INFO - Clicked Buy Medicines
2026-05-24 09:21:11,965 - INFO - Searched Medicine : Vicks Vaporub
2026-05-24 09:21:44,195 - INFO - ===== NEGATIVE SEARCH TEST =====
2026-05-24 09:21:46,533 - INFO - Popup Closed
2026-05-24 09:21:53,724 - INFO - Clicked Buy Medicines
2026-05-24 09:21:55,516 - INFO - Searched Invalid Medicine : xyzinvalidmedicine
2026-05-24 09:22:15,949 - INFO - Negative Search Test Passed
2026-05-24 09:22:25,591 - INFO - ===== ADD TO CART TEST =====
2026-05-24 09:22:30,183 - INFO - Clicked Buy Medicines
2026-05-24 09:22:32,219 - INFO - Searched Product : Dolo 650
2026-05-24 09:22:55,753 - INFO - Product Added To Cart
2026-05-24 09:23:02,254 - INFO - Cart Opened Successfully
2026-05-24 09:23:07,502 - INFO - Product In Cart : Product Added
2026-05-24 09:23:07,607 - INFO - Add To Cart Test Passed
2026-05-24 09:23:07,912 - INFO - Payment Process Started
2026-05-24 09:23:18,998 - INFO - ===== ADD TO CART TEST =====
2026-05-24 09:23:23,068 - INFO - Clicked Buy Medicines
2026-05-24 09:23:25,221 - INFO - Searched Product : Paracetamol 500mg
2026-05-24 09:23:48,143 - INFO - Product Added To Cart
2026-05-24 09:23:55,049 - INFO - Cart Opened Successfully
2026-05-24 09:24:00,411 - INFO - Product In Cart : Product Added
2026-05-24 09:24:00,526 - INFO - Add To Cart Test Passed
2026-05-24 09:24:00,831 - INFO - Payment Process Started
2026-05-24 09:24:11,801 - INFO - Starting Multi-Item End-to-End Test
2026-05-24 09:24:11,809 - INFO - Apollo 24/7 website launched successfully
2026-05-24 09:24:12,057 - INFO - Initiating Login sequence
2026-05-24 09:24:33,136 - INFO - Login completed successfully
2026-05-24 09:24:33,374 - INFO - Processing item 1: Dolo 650
2026-05-24 09:25:05,360 - INFO - Processing item 2: Paracetamol 500mg
2026-05-24 09:25:28,927 - INFO - Processing item 3: Vicks Vaporub
2026-05-24 09:25:53,430 - INFO - Opening cart to verify and checkout

2026-05-24 09:25:55,750 - INFO - Multi-item Checkout process initiated



The screenshot shows a web browser window with a project named 'Buy_Medicine_Module'. The left sidebar displays a file tree with folders like 'config', 'data', 'logs', 'pages', 'reports', 'tests', and 'utils'. The 'logs' folder is expanded, showing 'automation.log'. The main content area displays the log file's contents, which are timestamped and include various test steps and results. The log entries are as follows:

Line Number	Timestamp	Message
2672	2026-05-24 09:44:12,856	INFO - Initiating Login sequence
2673	2026-05-24 10:03:15,940	INFO - ===== WEBSITE LAUNCH TEST =====
2674	2026-05-24 10:03:16,213	INFO - Website Launch Test Passed
2675	2026-05-24 10:03:24,810	INFO - ===== POSITIVE LOGIN TEST =====
2676	2026-05-24 10:03:25,223	INFO - Clicked Login Button
2677	2026-05-24 10:03:26,067	INFO - Entered Mobile Number : 9876543210
2678	2026-05-24 10:03:26,398	INFO - Clicked Continue Button
2679	2026-05-24 10:03:26,771	INFO - Waiting For Manual OTP Entry
2680	2026-05-24 10:03:47,419	INFO - Clicked Verify Button
2681	2026-05-24 10:03:47,655	INFO - Positive Login Test Passed
2682	2026-05-24 10:03:55,590	INFO - ===== NEGATIVE LOGIN TEST =====
2683	2026-05-24 10:03:55,881	INFO - Clicked Login Button
2684	2026-05-24 10:03:56,788	INFO - Entered Invalid Mobile Number : 123456789
2685	2026-05-24 10:03:59,277	INFO - Negative Login Test Passed
2686	2026-05-24 10:04:08,664	INFO - ===== POSITIVE SEARCH TEST =====
2687	2026-05-24 10:04:10,923	INFO - Popup Closed
2688	2026-05-24 10:04:14,750	INFO - Clicked Buy Medicines
2689	2026-05-24 10:04:16,637	INFO - Searched Medicine : Dolo 650
2690	2026-05-24 10:04:19,842	INFO - Medicine Search Successful
2691	2026-05-24 10:04:20,097	INFO - Positive Search Test Passed
2692	2026-05-24 10:04:46,499	INFO - ===== POSITIVE SEARCH TEST =====
2693	2026-05-24 10:04:48,780	INFO - Popup Closed
2694	2026-05-24 10:04:52,795	INFO - Clicked Buy Medicines
2695	2026-05-24 10:04:54,756	INFO - Searched Medicine : Paracetamol 500mg
2696	2026-05-24 10:04:57,718	INFO - Medicine Search Successful
2697	2026-05-24 10:04:57,843	INFO - Positive Search Test Passed
2698	2026-05-24 10:05:06,377	INFO - ===== POSITIVE SEARCH TEST =====
2699	2026-05-24 10:05:08,680	INFO - Popup Closed
2700	2026-05-24 10:05:12,268	INFO - Clicked Buy Medicines
2701	2026-05-24 10:05:13,722	INFO - Searched Medicine : Vicks Vaporub
2702	2026-05-24 10:05:34,101	WARNING - Medicine Search Result Not Found
2703	2026-05-24 10:05:34,248	INFO - Positive Search Test Passed

Test Reports

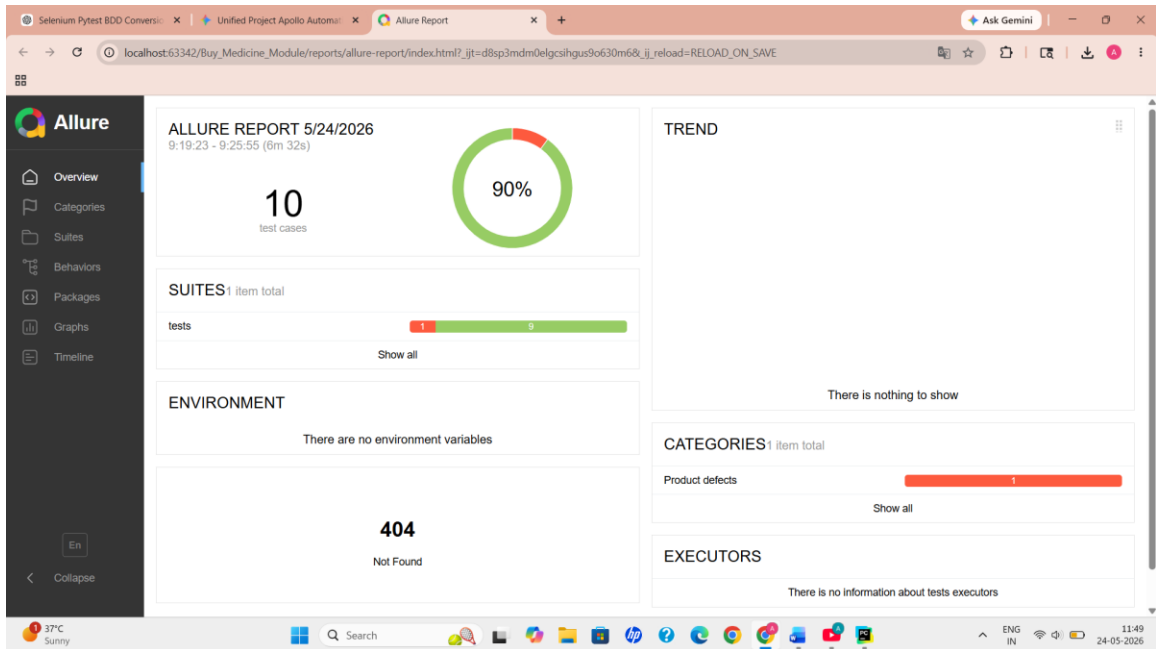
Allure Reporting Integration

The framework integrates Allure Reports to provide:

- Detailed execution summary
- Step-by-step execution details
- Pass/Fail statistics
- Screenshot attachments
- Execution timeline
- Scenario-based reporting

Allure Result Directory: reports/allure-results/

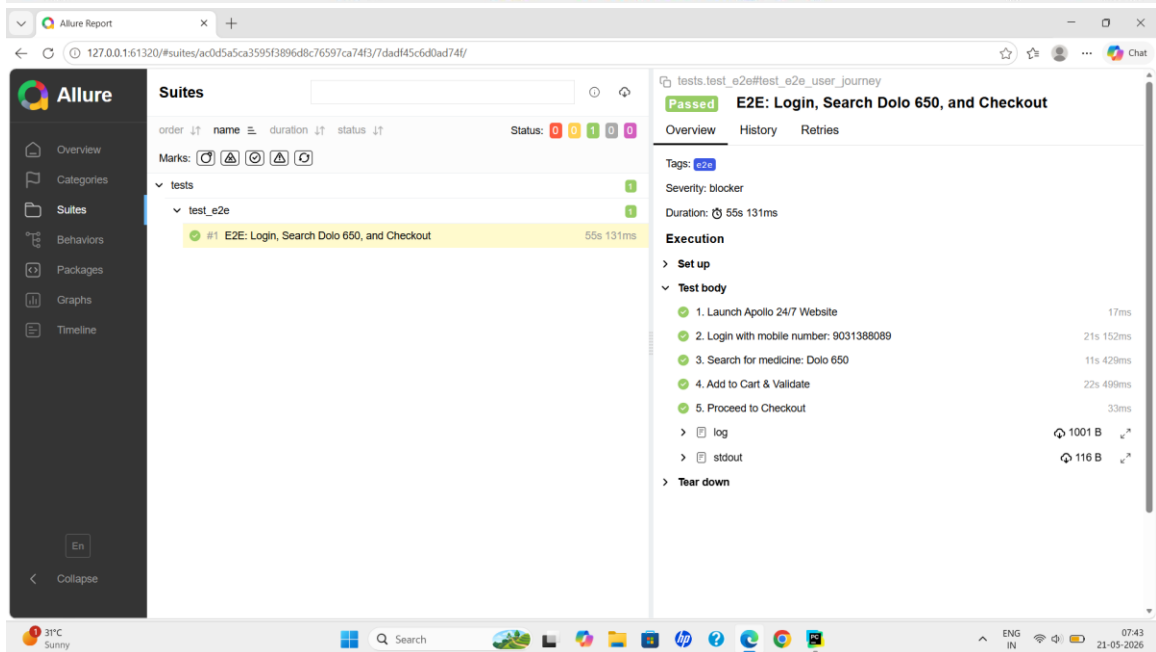
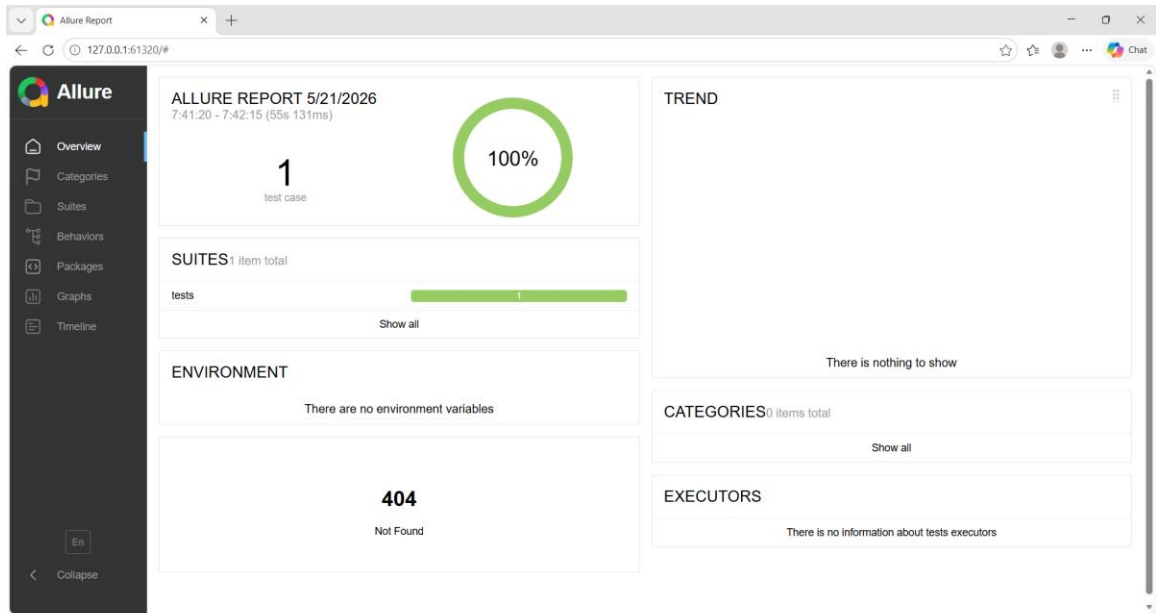
Generated HTML Report Directory: reports/allure-report/



The Allure Report Suites table displays the following information:

order	name	duration	status
Status: 1 0 0 0 0			
Marks: [Icons]			
▼ tests			
▼ test_all			
#8	Add To Cart Test - Dolo 650	{product_name: 'Dolo 650', 'expected': 'su...}	42s 504ms
#9	Add To Cart Test - Paracetamol 500mg	{product_name: 'Paracetamol 500mg', 'e...}	42s 062ms
#3	Negative Login Test - 123456789		3s 514ms
#7	Negative Search Test - xyzinvalidmedicine		31s 921ms
#2	Positive Login Test - 9876543210		22s 887ms
#4	Positive Search Test - Dolo 650	{search_text: 'Dolo 650', 'expected': 'su...}	11s 728ms
#5	Positive Search Test - Paracetamol 500mg	{search_text: 'Paracetamol 500mg', 'e...}	8s 941ms
#6	Positive Search Test - Vicks Vaporub	{search_text: 'Vicks Vaporub', 'expect...}	29s 166ms
#1	Verify Apollo Website Launch		954ms
▼ test_e2e			
#1	E2E: Login, Add Multiple Items, and Checkout		1m 44s

One test failed due to website delay.

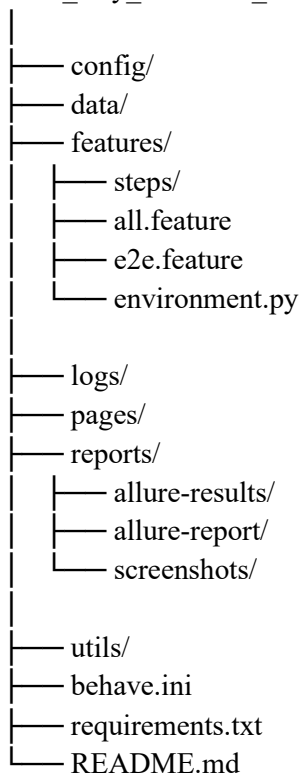


Framework Features

- Pure BDD Framework using Behave
- Selenium WebDriver Automation
- Page Object Model Architecture
- CSV Data-Driven Testing
- Allure Reporting
- Screenshot Capture
- Logging Mechanism
- End-to-End Automation
- Positive & Negative Testing
- Reusable Components
- Scalable Framework Structure

Folder Structure

BDD_Buy_Medicine_Module/



Buy_Medicine_Module (Selenium)/

|— config/

| |— config.properties

|— data/

| |— login_data.csv

| |— medicine_search_data.csv

| |— product_validation_data.csv

|— logs/

| |— automation.log

|— pages/

| |— base_page.py

| |— cart_page.py

| |— login_page.py

| |— product_page.py

| |— search_medicine_page.py

|— reports/

| |----- allure-report/

| |— allure-results/

| |— screenshots/

|— tests/

| |— test_all.py

| |— test_e2e.py

|— utils/

| |— config_reader.py

| |— csv_utils.py

| |— driver_factory.py

- | | — excel_utils.py
- | | — logger.py
- | | — screenshot_utils.py
- | — conftest.py
- | — pytest.ini
- | — requirements.txt

Conclusion

The Apollo 24/7 Buy Medicine Automation Framework successfully automates major business-critical workflows of the Apollo 24/7 application using Selenium WebDriver with Python and Behavior Driven Development (BDD) methodology.

The framework effectively validates functionalities such as website launch, user authentication, medicine search, add to cart operations, and checkout workflow through both positive and negative test scenarios. The implementation of the Page Object Model (POM) design pattern improves code reusability, maintainability, and scalability.

Integration of Behave for BDD execution, Allure Reports for advanced reporting, CSV-based data-driven testing, screenshot capturing, and logging mechanisms makes the framework highly organized and efficient for real-world automation testing requirements.

The project demonstrates practical implementation of modern automation testing concepts and provides a scalable foundation for enterprise-level web automation testing and continuous testing practices.